Cover letter to the Editors of *Physical Review A*

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October 6, 2026

Dear Editors,

On October 6, 2026, we submitted our manuscript "**Blind catalytic quantum error correction: target-state estimation and fidelity bounds**", for consideration as a Regular Article in *Physical Review A*. The submission consists of the main manuscript (17 figures, 9 tables) and a frozen release of the open-source package `blind_cqec` v0.2.0 (MIT license, https://github.com/deeptell-inc/blind_cqec_pkg) with a 91-test reproducibility suite, shipped raw results files, and a script-to-figure map.

What we discovered. Catalytic quantum error correction (CQEC) is a threshold-free state-recovery framework whose original formulation requires the ideal target state to be known exactly. In this study we show that the target-knowledge requirement can be displaced onto a classical estimation problem. Under an explicitly stated recovery interface—the mode-inclusion-restricted projection of the estimate—recovery fidelity is bounded by estimation fidelity plus a mode-survival penalty, and an 84-condition benchmark realizes both regimes of that bound. The resulting estimator design-space map is sharp: with exactly calibrated noise, channel inversion coincides with the oracle at every dimension and both are capped by a mode-survival ceiling; the noise-model-free alternative, coherence maximization, matches the ceiling to within 1.5% at $d \leq 16$ and degrades with dimension; and the operational choice between the two is decided by calibration accuracy, with the dephasing rate exponentially dominant.

Why this matters. The result separates, for the CQEC setting, what is a classical estimation question from what is a property of the recovery interface, and it reports the negative results with the same prominence as the positive ones: coherence maximization is actively harmful for mixed targets, no strategy reaches high fidelity where mode annihilation sets the ceiling, and universal single-copy recovery is impossible for any deterministic target-independent apparatus. This map is the prerequisite for the two open problems the manuscript states as future work—an explicit joint covariant recovery channel and an operational measurement model—and it is directly useful to practitioners choosing an estimator for variational and iterative algorithms whose ideal output is unknown to the correction module.

Why *Physical Review A*. The manuscript combines a resource-theoretic result (a one-sided fidelity bound for the mode-restricted recovery interface, derived step by step from the Shiraishi–Takagi catalytic theorem and the Jordan–Hahn decomposition) with a systematic numerical study of estimation strategies, and it situates both against the recent literature on quantum error mitigation, catalysis, and the sample complexity of state learning. This combination of quantum information theory and numerical benchmarking of near-term error handling is the core of the journal's quantum information section. The work is fully reproducible: every figure and table is generated by a released script with its raw data shipped alongside, and the claims are scoped to what is demonstrated, with the limitations stated in the abstract, methods, and a dedicated section.

Suggested referees.

1. Ryuji Takagi, Nanyang Technological University, Singapore (`ryuji.takagi@ntu.edu.sg`) — catalytic coherence transformations.

2. Mark M. Wilde, Cornell University, USA (wilde@cornell.edu) — quantum Shannon theory, channel fidelity bounds.
3. Andreas Winter, Universitat Autònoma de Barcelona, Spain (andreas.winter@uab.cat) — catalytic resource theories.
4. Bálint Koczor, University of Oxford, UK (balint.koczor@maths.ox.ac.uk) — error mitigation, near-term VQE.
5. Daniel Stilck França, ENS Lyon / CNRS, France (daniel.stilck-franca@ens-lyon.fr) — quantum state estimation, copy complexity.
6. Iman Marvian, Duke University, USA (iman.marvian@duke.edu) — asymmetry resource theory, covariant operations.

Non-preferred referees. None.

Statements. This manuscript is not under consideration elsewhere. An earlier version was reviewed at another journal and declined; the present version incorporates the resulting corrections, in particular the explicit statement that the catalytic transformation itself is not simulated, that all resource counts are conditional on oracle density-matrix access, and that the high-dimensional recovery ceilings depend on the numerical mode threshold (now tabulated). A preprint is available as arXiv:2604.11857. The author has read and approved the submission and declares no competing interests.

Sincerely,

Hikaru Wakaura
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